

4-Channel ADC Module Prototype

Module Description and Analysis

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1 Introduction

This document describes the schematic and functional operation of a 4-channel ADC module prototype, designed in KiCad 9. The module is one building block in a scalable detector-controller system: each board contains its own processor and plugs into a minimal backplane that provides:

- A single 12 V power rail.
- A 10 MHz synchronization clock for the DC-DC converters and the detector sequencer
- A start signal indicating the beginning of an acquisition cycle.
- An emergency-stop line to flag a fault condition.

Each board also has its own Gigabit Ethernet port—for configuration, telemetry, and streaming video data

Figure 1 and Figure 2 show the PCB implementation of the 4-channel ADC prototype board with

- | | |
|--------------------------------------|--|
| 1. Digital 3.3 V DC-DC Converter. | 12. Digital 2.5 V LDO. |
| 2. Backplane Connector. | 13. 4 Channel Video Front-End and ADC. |
| 3. Analog -15.5 V DC-DC Converter. | 14. Analog 5 V LDO. |
| 4. RJ45 Ethernet Connector. | 15. Analog 2.5 V LDO. |
| 5. Zynq-Soc Module Mode Switch. | 16. Analog 4.2 V LDO. |
| 6. Voltage Translator IC. | 17. Analog 15 V LDO. |
| 7. Analog 5.5 V DC-DC Converter. | 18. Analog -15 V LDO. |
| 8. Analog 15.5 V DC-DC Converter. | 19. Video Input Connector. |
| 9. FTDI USB-C Interface. | 20. 12 V Power Input Protection. |
| 10. Zynq-Soc Module. | 21. Clock Generator. |
| 11. Zynq-Soc Reset Button. | 22. EEPROM. |

In Figure 1, most of the digital circuitry is clustered on the left side (from top view). Along the top edge you'll find the backplane connector and the DC-DC converters; along the bottom edge are the low-noise LDOs, the four ADC channels with their analog front end, and the input connector.

Immediately beside the top backplane connector, a jitter-attenuator clock generator takes the 10 MHz sync clock from the backplane and delivers a low-jitter reference to the Zynq SoC for the detector sequencer and DC-DC synchronization. A small I²C EEPROM stores the board's type and revision information.

At the center of the board sits a Trenz TE0720-04-62I33MA Zynq SoC-Module, which integrates an AMD Zynq™ 7020-2I, 1 GB of DDR3L RAM, and 8 GB of eMMC flash.

On the left digital side, a Gigabit-Ethernet RJ45 port interfaces with the Zynq SoC for configuration, telemetry, and streaming video data from the ADCs. An FTDI USB-C interface provides direct programming of the Zynq (without a separate JTAG adapter) and supplies a UART console link

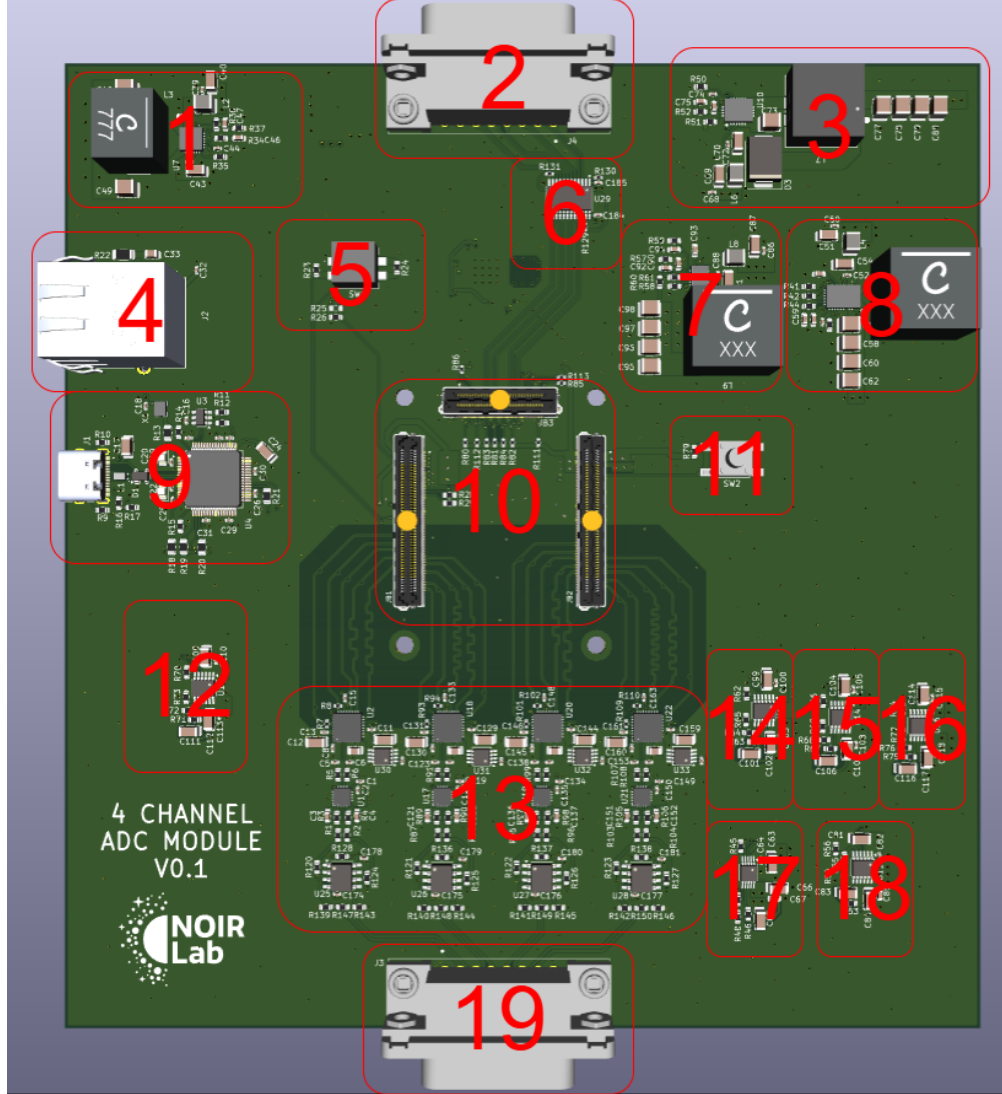


Figure 1: Top View of the 16.2 cm x 17.1 cm, six-layer PCB with impedance control, 4-channel ADC prototype board.

2 Power

2.1 12 V Input Protection

As 12 V input protection we use a TPS259480AYWPR Texas Instruments eFuse that features:

To safeguard the 12 V supply, we employ a Texas Instruments TPS259480A e-Fuse, which integrates back-to-back MOSFETs and offers the following features:

- Integrated back-to-back FETs with low ON resistance $R_{ON} = 12.2 \text{ m}\Omega$.
- Ideal diode operation with true reverse current blocking.
- Fast overload protection.
- Overcurrent protection with load current monitor.

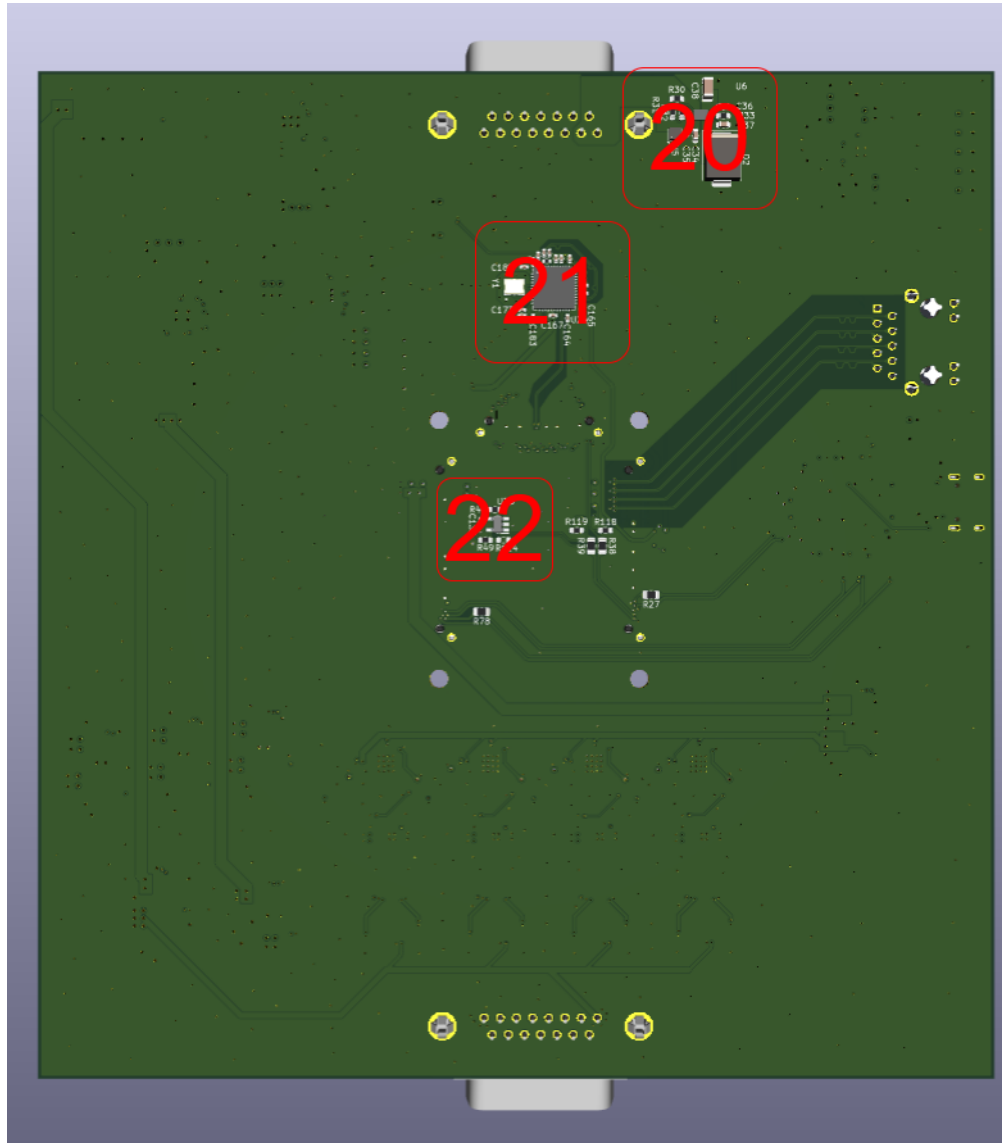


Figure 2: Bottom View of the 16.2 cm x 17.1 cm, six-layer PCB with impedance control, 4-channel ADC prototype board.

- Fast-trip response for short circuit protection.
- Adjustable output slew rate.

Upstream of the e-Fuse, a TI TVS1800DRV 18 V flat-clamp transient-voltage suppressor absorbs voltage spikes and surges on the 12 V line. Downstream, a B520C 20 V, 5 A Schottky diode provides secondary overvoltage protection at the e-Fuse output. Figure 3 illustrate the complete 12 V input protection circuit.

2.2 Digital 3.3 V (3V3D)

Digital 3.3 V serves as the primary supply for the Zynq SoC and feeds the 2.5 V digital LDO used by the ADC. We selected the Analog Devices LT8642S synchronous step-down converter, which provides:

- Ultra low EMI emissions (Silent Switcher 2 Architecture).

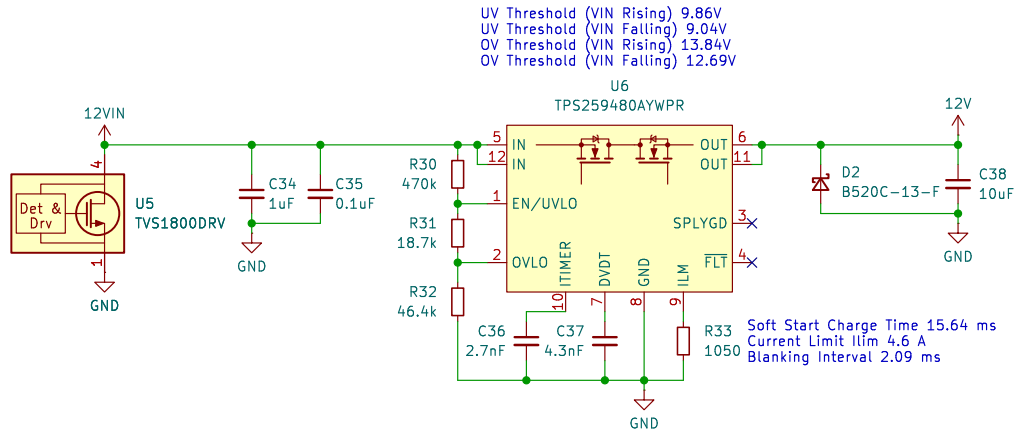


Figure 3: 12 V Input Protection Circuit

- High efficiency: up to 95% efficiency at 2 MHz, 12 V input and 3.3 V output.
- Up to 10 A output current.
- External compensation.

The LT8642S is configured to operate at 2 MHz (see Figure 4). A Coilcraft XGL1010-801 shielded inductor (0.8 μ H, 0.88 m Ω DCR) minimizes losses. Upstream of the converter input, a π LC filter suppresses input noise. Power-Good output (3V3D_PG) and the SYNC pin (3V3D_SYNC) are both tied back to the Zynq SoC for voltage monitoring and clock synchronization. See LTSpice and LTPoweCAD simulation files for transient and frequency compensation design.

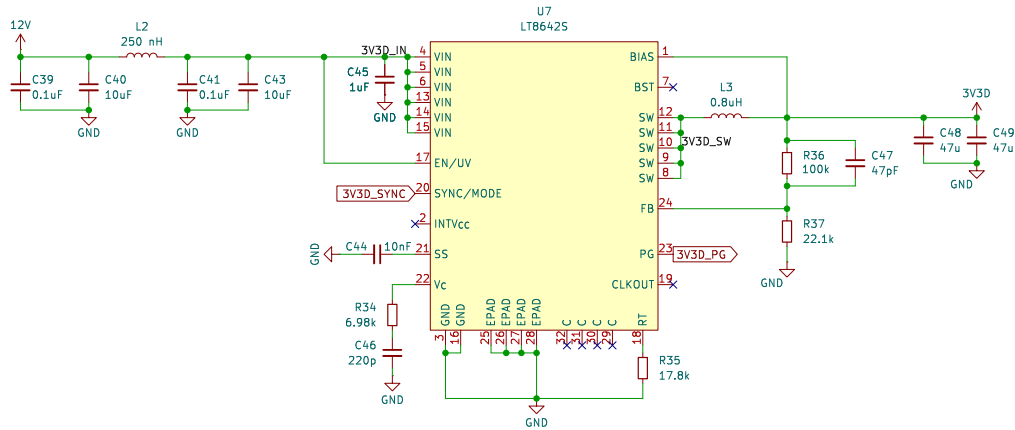


Figure 4: Digital 3.3 V Step-Down DC-DC Converter Circuit

2.3 Analog 5.5 V (5V5A)

Analog 5.5 V (5V5A) powers low noise analog LDO that supply analog front end and ADC. We selected the Analog Devices LT8624S synchronous step-down DC-DC converter that features:

- Ultralow 4 nV/ $\sqrt{\text{Hz}}$ RMS noise (Silent Switcher 3 architecture).
- High efficiency.
- Ultra fast transient response: 1 μ s.

- Up to 4 A continuous output current.
- External compensation.

The LT8624S is configured to operate at 2 MHz (see Figure 5). A Coilcraft XGL1313-222 shielded inductor (2.2 μ H, 1.4 m Ω DCR) minimizes losses. Upstream of the converter input, a π LC filter suppresses input noise. Enable (5V5A_EN), Power-Good output (5V5A_PG) and the SYNC pin (5V5A_SYNC) are connected to the Zynq SoC for enabling, voltage monitoring and clock synchronization. See LTSpice and LTPoweCAD simulation files for transient and frequency compensation design.

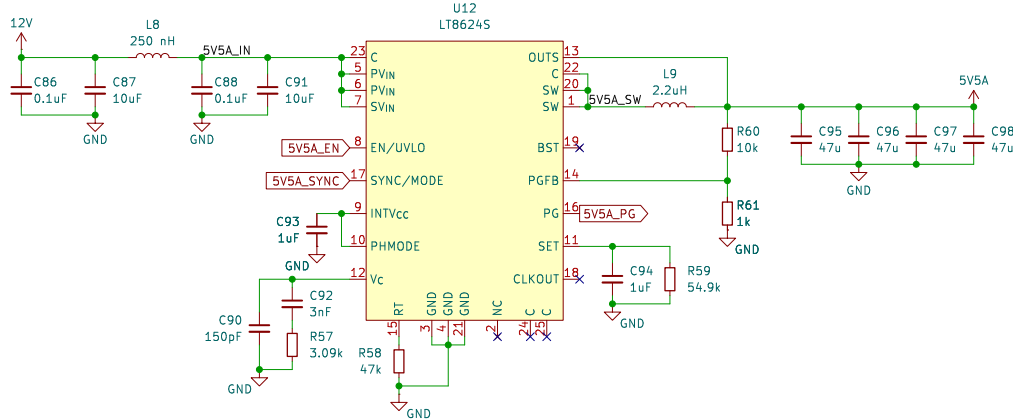


Figure 5: Analog 5.5 V Step-Down DC-DC Converter Circuit

2.4 Analog 15.5 V (+15V5A)

The Analog 15.5 V (+15V5A) DC-DC converter powers the low-noise Analog 15 V LDO (15VA). We chose the Analog Devices LT8350S synchronous buck-boost converter that features:

- 4-switch single inductor architecture.
- Low EMI (Silent Switcher 2 architecture).
- Up to 95% efficiency at 2 MHz.
- $\pm 1.5\%$ output voltage regulation.
- External compensation.

The LT8350S is configured to operate at 2 MHz (see Figure 6). A Coilcraft XGL1313-222 shielded inductor (2.2 μ H, 1.4 m Ω DCR) minimizes losses. Upstream of the converter input, a π LC filter suppresses input noise. Enable (15V5A_EN), Power-Good output (15V5A_PG) and the SYNC pin (15V5A_SYNC) are connected to the Zynq SoC for enabling, voltage monitoring and clock synchronization. See LTSpice and LTPoweCAD simulation files for transient and frequency compensation design.

2.5 Analog -15.5 V (-15V5A)

Analog -15.5 V (-15V5A) DC-DC converter powers the low-noise -15 V LDO (-15VA). We choose the Analog Devices LT3579 synchronous boost-inverting DC-DC converter that features:

- 6 A, 42 V combined power switch.
- Output short circuit protection.

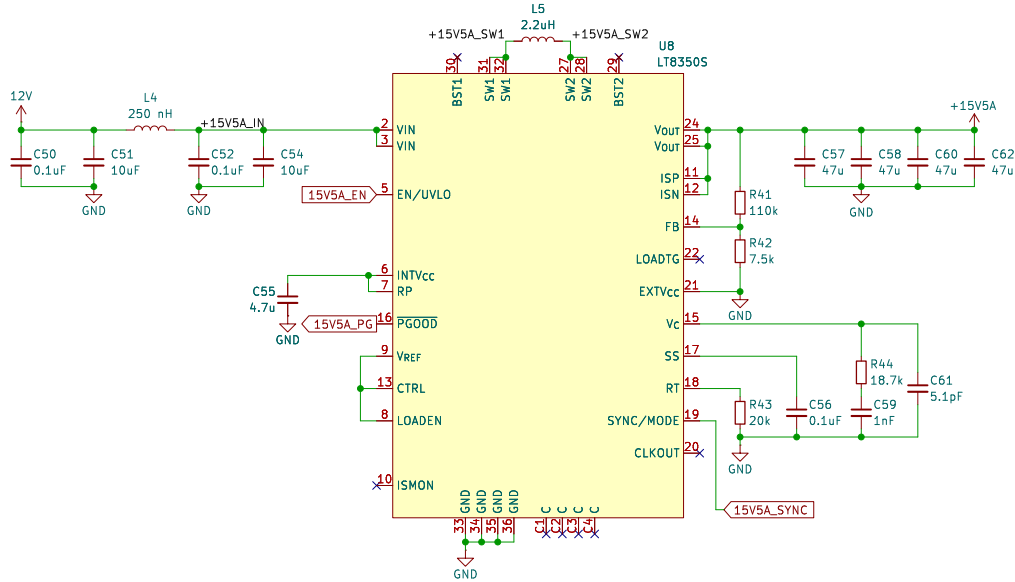


Figure 6: Analog 15.5 V Buck-Boost DC-DC Converter Circuit

- Configurable as inverting converter.
- External compensation.

The LT3579 is configured to operate at 2 MHz (see Figure 7). An Eaton DRQ-125-3R3-R coupled inductor (3.3 μ H, 6.3 m Ω DCR) minimizes losses. Upstream of the converter input, a π LC filter suppresses input noise. Enable (-15V5A_EN), Power-Good output (-15V5A_PG) and the SYNC pin (-15V5A_SYNC) are connected to the Zynq SoC for enabling, voltage monitoring and clock synchronization. See LTSpice for transient design.

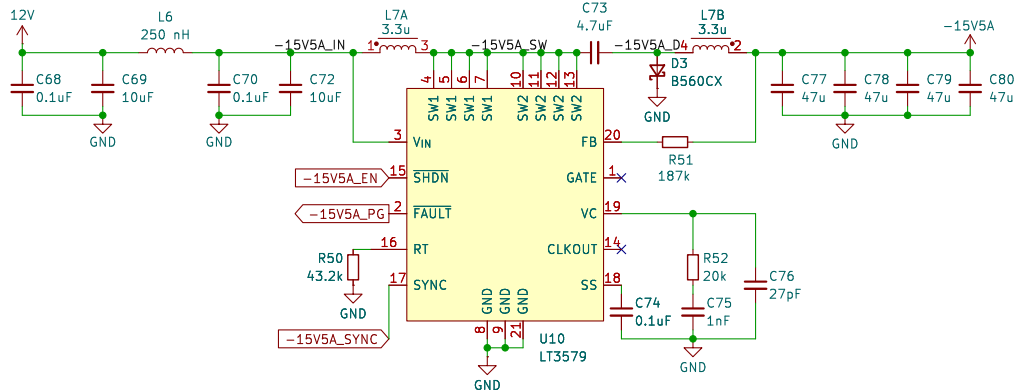


Figure 7: Analog -15.5 V Inverting DC-DC Converter Circuit

2.6 Linear Regulators

The board requires multiple low-noise linear regulators to generate the analog and digital rails from the protected input supply. The following voltages are provided:

- Analog Supplies: 15 V (15VA), -15 V (-15VA), 5 V (5VA), 4.2 V (4V2A) and 2.5 V (2V5A) for the analog front-end, ADCs and any external circuitry.

- Digital Supplies: 2.5 V (2V5D) for the ADC LVDS and the Zynq SoC LVDS I/O bank.

All rails are generated by Analog Devices LT3042 (15VA), LT3093 (-15VA) and LT3045 (5VA, 4V2A, 2V5A and 2V5D) ultra-low noise LDOs. These regulators offers:

- Extremely low output noise: 0.8 μV RMS (10 Hz to 100 kHz) and 2 nV/ $\sqrt{\text{Hz}}$ at 10 kHz.
- Ultrahigh PSRR: 73 dB (LT3093), 76 dB (LT3045) and 79 dB (LT3042) at 1 MHz.
- 500 mA (LT3045) and 200 mA (LT3093 and LT3042) output current.

Figure 8 shows the schematic of the analog 5 V regulator. Aside from the different resistor values used for setting the output voltage and the power-good feedback network, all LDO circuits follow the same layout. Each device's Enable (EN) and Power-Good (PG) pins are connected back to the Zynq SoC for centralized sequencing and fault monitoring.

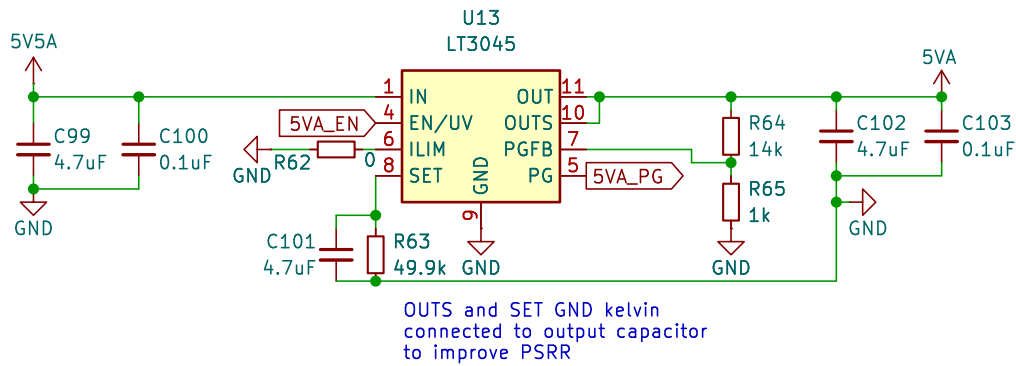


Figure 8: Analog 5 V LDO Circuit

3 Clock Generator

The clock generator derives an ultra-low-jitter timing reference from the single-ended 10 MHz backplane synchronization signal (CLOCK_SYNC). We use the Skyworks Si5342A-D jitter-attenuating clock multiplier, which provides:

- Ultralow 90 fs RMS output jitter.
- Support for external crystals between 25 MHz and 54 MHz.
- SPI/I²C control interface.
- Zero-delay mode for minimal, deterministic input-to-output latency.
- Both single-ended and differential I/O.

The Si5342A-D “cleans” the backplane clock to produce a low-jitter reference. The Zynq SoC then employs this cleaned clock for its sequencer timing and to drive the DC-DC converters, reducing their switching noise and improving overall power-rail integrity.

In our design (see Figure 9), a Kyocera AVX 48 MHz $\pm$ 10ppm crystal provides the Si5342A-D’s reference frequency. The device is configured in zero-delay mode to guarantee a fixed, repeatable phase relationship between CLOCK_SYNC and the output clock to Zynq SoC. Configuration and status monitoring are handled over I²C via the Zynq SoC’s. Finally, the Si5342A-D’s differential 2.5 V LVDS output is routed directly into one of the Zynq SoC’s MRCC (Multi-Region Clock-Capable) input.

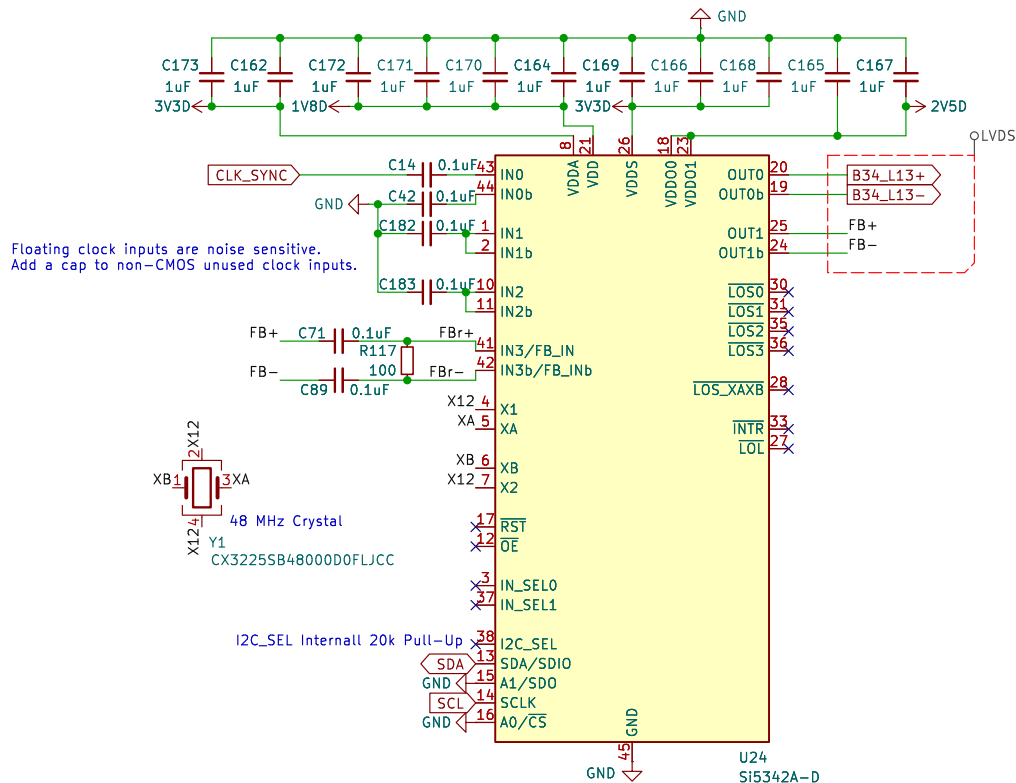


Figure 9: Si5342A-D Clock Generator Jitter Attenuator Circuit

4 Analog Front-End and ADC

The analog front-end of the 4-channel ADC module is organized as two-stage chain that conditions each video input before digitalization. Figure 10 illustrates the complete signal path for a single video channel. First, a low-noise instrumentation amplifier provides a well-defined return path, input impedance and gain for the input differential signal from detector. Second, a Fully Differential amplifier implements a low-pass filter and sets common-mode voltage for ADC. And finally, a high-speed SAR ADC samples the video signal and send data to Zynq SoC through 2.5 V LVDS.

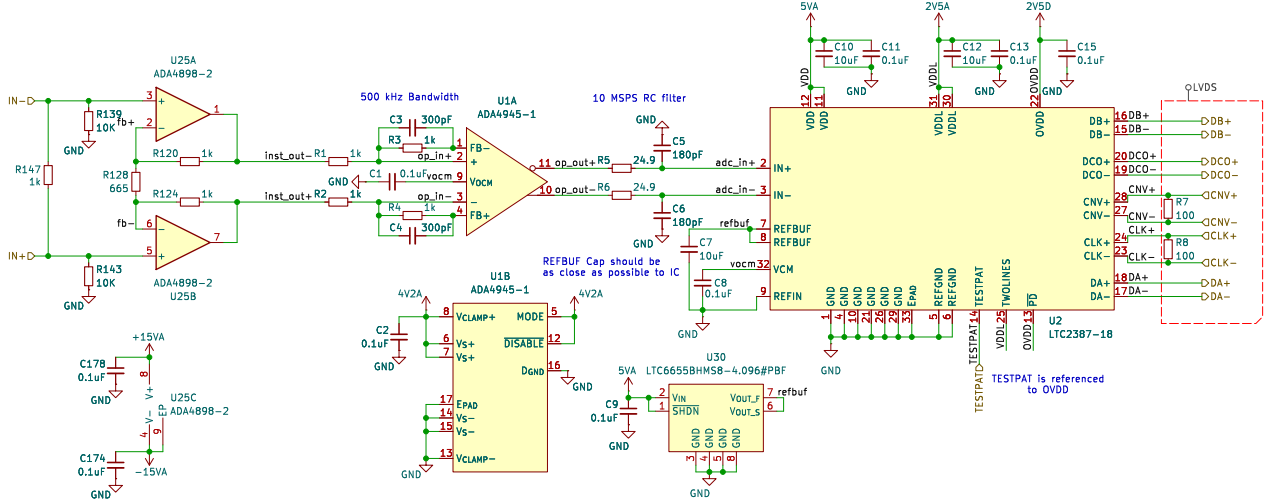


Figure 10: Analog Front-End and ADC Circuit.

Using Analog Devices Signal Chain Designer, the total noise referred to the input from 0.1 Hz to 5 MHz is $12.9 \mu\text{V}_{\text{rms}}$, where:

- Instrumentation amplifier: $2.94 \mu\text{V}_{\text{rms}}$.
- Fully differential amplifier: $3.52 \mu\text{V}_{\text{rms}}$.
- ADC: $12 \mu\text{V}_{\text{rms}}$.

Referred to front-end input:

- Input signal swing: $v_{\text{in}}^{\text{max}} = 8.192/4 = 2.05 \text{ V}$ (DC gain of 4 V/V and ADC input range of 8.192 V).
- AC max signal: $v_{\text{in,AC}}^{\text{max}} = v_{\text{in}}^{\text{max}}/2/\sqrt{2} = 723 \text{ mV}_{\text{rms}}$.
- Signal chain noise: $n_{\text{in}} = 12.9 \mu\text{V}_{\text{rms}}$.
- The Signal-to-Noise ratio (SNR): $20 \log_{10}(v_{\text{in,AC}}^{\text{max}}/n_{\text{in}}) = 94.99 \text{ dB}$.
- Effective-Number-of-Bits (ENOB) without distortion: $(\text{SNR} - 1.76)/6.02 = 15.49 \text{ bit}$.

4.1 Instrumentation Amplifier

Figure 11 shows the two opamp instrumentation amplifier equivalent circuit. Resistor $R_1 = 1 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$ define input impedance, common mode path and return path for video input. Output common-mode voltage is maintained and differential gain G_1 is set by $R_3 = 665 \Omega$ and $R_4 = 1 \text{ k}\Omega \pm 0.1 \%$ as:

$$G_1 \equiv \frac{v_{out}}{v_{in}} = \frac{v_{out}^+ - v_{out}^-}{v_{in}^+ - v_{in}^-} = 1 + \frac{2R_4}{R_3} \approx 4 \text{ V/V} \quad (1)$$

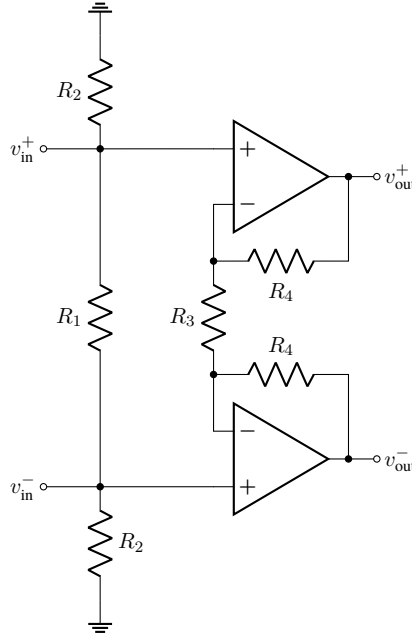


Figure 11: Instrumentation Amplifier Equivalent Circuit.

For instrumentation amplifier we selected the Analog Devices ADA4898-2 dual operational amplifier that features:

- Ultralow voltage noise: $0.9 \text{ nV}/\sqrt{\text{Hz}}$ and $2.4 \text{ pA}/\sqrt{\text{Hz}}$ at 1 kHz.
- Ultralow distortion: -93 dBc at 500 kHz.
- High speed: 65 MHz bandwidth at unity gain.
- Unity gain stable.
- Low input offset: $160 \mu\text{V}$ maximum.

4.2 Fully Differential Amplifier

Figure 12 shows the fully differential circuit equivalent circuit. Resistors $R_1 = R_2 = 1 \text{ k}\Omega \pm 0.1 \%$ define input impedance and differential gain. Output common-mode voltage is set externally by ADC to 2.048 V and differential gain G_2 is set as:

$$G_2 \equiv \frac{v_{out}}{v_{in}} = \frac{v_{out}^+ - v_{out}^-}{v_{in}^+ - v_{in}^-} = \frac{R_2}{R_1} = 1 \text{ V/V} \quad (2)$$

The bandwidth BW is defined by R_2 and $C_1 = 300 \text{ pF}$ for 1 MSPS as:

$$\text{BW} = \frac{1}{2\pi R_2 C_1} \approx 530 \text{ kHz} \quad (3)$$

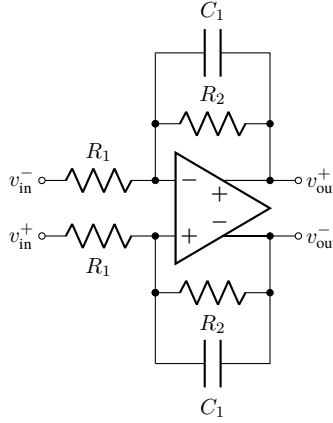


Figure 12: Fully Differential Amplifier Equivalent Circuit.

As fully differential amplifier we selected the Analog Devices ADA4945-1 in full-power mode that features:

- Rail-to-rail output.
- 4 mA at full-power mode.
- 145 MHz bandwidth at gain 1 at full-power mode.
- Low noise: $2 \text{ nV}/\sqrt{\text{Hz}}$ at 100 kHz.
- 115 μV maximum offset voltage.

4.3 Analog-to-Digital Converter (ADC)

Figure 13 shows ADC input filter equivalent circuit. For an 18 bit, the error of a full scale range (FS) step in a single-pole RC filter after acquisition time t_{ACQ} is $\exp\left(-\frac{t_{\text{ACQ}}}{R_{\text{filt}}C_{\text{filt}}}\right)$ FS and a half LSB is approximately 2^{-19} FS. Then we have

$$\exp\left(-\frac{t_{\text{ACQ}}}{R_{\text{filt}}C_{\text{filt}}}\right) \text{FS} = 2^{-19} \text{FS} \Leftrightarrow C_{\text{filt}} = \frac{t_{\text{ACQ}}}{19R_{\text{filt}} \ln 2} \quad (4)$$

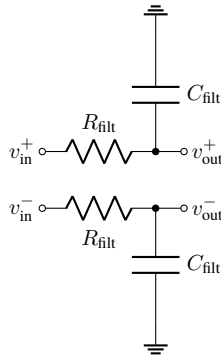


Figure 13: ADC Input filter Equivalent Circuit.

We selected the Analog Devices LTC2387-18 18 bit 15 MSPS SAR ADC that features:

- No pipeline delay, no cycle latency.
- 95.7 dB SNR at 1 MHz.
- 102 dB SFDR at 1 MHz.
- ± 3 LSB INL maximum.
- 8.192 V_{p-p} differential input.
- 2.5 V Serial LVDS interface.

For selected ADC we have $t_{ACQ} = 1/f_s - 39$ ns. For $f_s = 10$ MSPS sample rate and $R_{filt} = 24.9 \Omega$ we get $C_{filt} \approx 180$ pF.

To improve ADC reference voltage temperature drift, reference out pin (REFBUF) is externally driven by Analog Devices LTC6655 4.096 V low drift voltage precision reference that features:

- Low Noise: 0.25 ppm_{p-p} (0.1 Hz to 10 Hz).
- Low drift: 2 ppm/°C.
- High accuracy: ± 0.025 % maximum.
- Long term drift: 20 ppm/ $\sqrt{\text{kh}}$.

5 Main Processor

The main processor is in charge of generating the sequencer signals for reading the detector, configuring integrated circuits, monitoring interlocks, and communicating via Ethernet protocol. As main processor we selected the Trenz Electronic TE0720-04-62I33MA SoC-Module. Figure 14 shows TE0720-04-62I33MA SoC-Module.

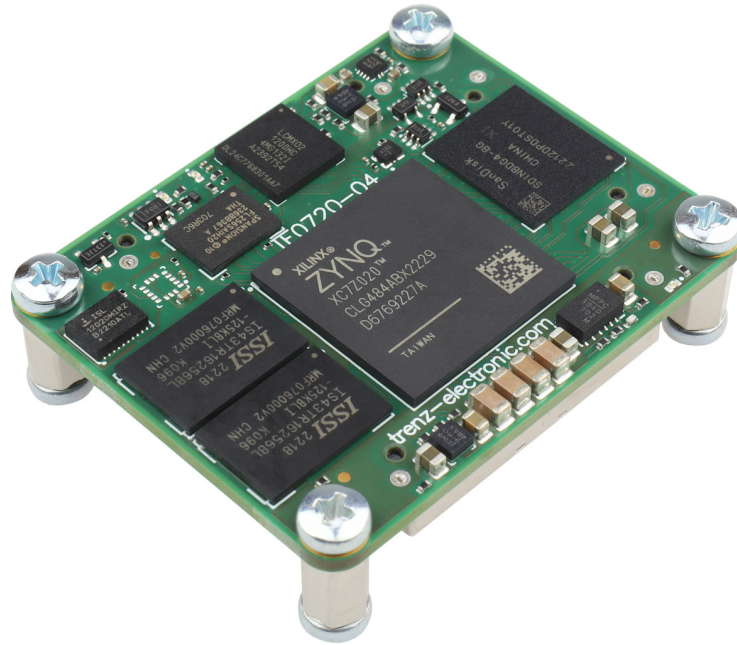


Figure 14: Trenz Electronic TE0720-04-62I33MA SoC-Module.

Trenz Electronic TE0720-04-62I33MA SoC-Module features:

- AMD Zynq™ XC7Z020-2CLG484I dual ARM® Cortex®-A9 MPCore™ with CoreSight™ System On Chip (SOC) IC Zynq®-7000 Artix™-7 FPGA, 85K Logic Cells 766MHz 484-CSPBGA (19x19).
- Rugged for high shock and vibration.
- 10/100/1000 tri-speed Gigabit Ethernet transceiver (PHY), SGMII accessible on a board-to-board connector.
- 1 GByte (32-bit) DDR3 SDRAM.
- 32 MByte QSPI Flash memory (for configuration and operation).
- 8 GByte eMMC.
- Plug-on module with 2 x 100-pin and 1 x 60-pin Razor Beam High-Speed hermaphroditic Terminal/Socket Strips.
- 152 FPGA I/O's (75 LVDS pairs possible) and 14 MIOs available on board-to-board connectors.
- On-board high-efficiency DC-DC converters.
- Programmable through Free Vivado Standard Edition Suite.

Trenz Electronic TE0720-04-62I33MA SoC-Module is configured as:

- Low jitter clock input for sequencer and DC-DC synchronization from clock generator IC.
- 2.5 V external power supply for banks 13, 34 and 35. Bank 13 is not powered (unused).
- JTAG programming and UART communication through on board FTDI.
- 3.3 V I2C communication through MIO pins for clock generator and EEPROM.
- 2.5 V LVDS differential pair communication with ADC.
- Internal 1.8 V is used for clock generator.
- Backplane communication through 2.5 V to 3.3 V voltage translator IC.

6 Connectors

6.1 Backplane Connector

Backplane connector serves as the primary interface for power, control, and monitoring systems. It brings in an external 12 V power rail to feed board DC-DC converters and 3.3 V digital I/O like synchronization and interlocks.

Selected backplane connector is TE Connectivity 5747841-4 15-position D-Sub plug male pins connector. Table 1 shows backplane connector pinout.

Pin name	Pin no.	Dir.	Level	Description
12VIN	7, 8, 15	—	12 V	External power input
BPI0	1	In	3.3 V	10 MHz external sync clock
BPI1	9	In	3.3 V	Digital input
BPI2	10	In	3.3 V	Digital input
BPI3	3	In	3.3 V	Digital input
BPO0	4	Out	3.3 V	Digital output
BPO1	12	Out	3.3 V	Digital output
BPO2	13	Out	3.3 V	Digital output
BPO3	6	Out	3.3 V	Digital output
GND	0, 2, 5, 11, 14	—	0 V	Ground return

Table 1: Backplane Connector Pinout

Backplane connector digital I/O signals are translated from 3.3 V to 2.5 V and then connected to 2.5 V Zynq SoC-Module bank 34. For voltage translation we selected a Texas Instrument SN74AXC8T245QPWRQ1 automotive 8 bit 380 Mbps dual-supply bus transceiver with ESD protection.

6.2 USB-C Connector

USB-C connector is used to program through JTAG and communicate through USART with Zynq SoC-Module using a USB converter.

Selected USB converter is FTDI FT2232HL USB Hi-Speed to Dual Channel Serial UART/FIFO/JTAG/SPI/I2C IC. FT2232HL ADBUS is used for JTAG and BDBUS is used for UART. UART RXD (Zynq to host) is connected to Zynq MI015 and TXD (host to Zynq) is connected to Zynq MI014.

6.3 Gigabit Ethernet Connector

Gigabit ethernet connector is RJ45 connector used for communication between Zynq SoC-Module Ethernet PHY and host for configuration, telemetry and video streaming.

Selected Gigabit ethernet connector is Taoglas TMJG4820GENL industrial grade RJ45 connector with integrated magnetics 1G Base-T that features EMI finger and short body. Internal LEDs are disabled.

6.4 Video Input Connector

Video input connector contains 4 differential video input pairs and also provides ± 15 V analog input to power external pre-amplifiers.

Selected video input connector is TE Connectivity 5747841-4 15-position D-Sub plug male pins connector. Table 2 shows video input connector pinout.

Pin name	Pin no.	Dir.	Level	Description
+15VA	1	Out	15 V	Analog power output
-15VA	8	Out	-15 V	Analog power output
IN0+	10	In	± 15 V	Channel 0 Video Input +
IN0-	2	In	± 15 V	Channel 0 Video Input -
IN1+	4	In	± 15 V	Channel 1 Video Input +
IN1-	11	In	± 15 V	Channel 1 Video Input -
IN2+	13	In	± 15 V	Channel 2 Video Input +
IN2-	5	In	± 15 V	Channel 2 Video Input -
IN3+	7	In	± 15 V	Channel 3 Video Input +
IN3-	14	In	± 15 V	Channel 3 Video Input -
GND	3, 6, 9, 12, 15	—	0 V	Ground return

Table 2: Video Input Connector Pinout

7 Miscellaneous

7.1 EEPROM

An EEPROM (Electrically Erasable Programmable Read-Only Memory) is used to store PCB board ID and model.

Selected EEPROM is Microchip 24LC16BT-I/OT 16 kbit non-volatile I2C EEPROM.